

Supplement 1: Empirical Findings Notebook

Joseph Corneli

Oxford Brookes University, Oxford OX3 0BP, United Kingdom

Hyperreal Enterprises Ltd, United Kingdom

Supplement to *Rule-Rewriting Cellular Automata and the Edge of Chaos*. Correspondence:

jcorneli@brookes.ac.uk.

Abstract

This notebook collects the twelve empirical findings referenced from the main paper. The numbering is stable: “Supplement 1, Box n ” in the paper points to Box n here.

Mathematical definitions, protocols, figures, and interpretation remain in the main paper; this supplement isolates the compact claims that those materials support.

How to read this supplement

The boxes below occur in the same order as the corresponding results in the main paper. References to figures, tables, theorems, and sections without a “Supplement” prefix refer to the main paper. The source file `supplement1-findings.tex` is canonical: the main paper contains pointers, not duplicate finding text.

Box 1: The full permutation census

Every permutation containing an odd cycle lacks a fixed byte (Supplement 2, Theorem 1); under the census protocol, all 29,295 such operators were classified genotype-alive. The theorem excludes a common fixed byte for all elementary writes; the observed 100% alive rate is an additional dynamical result. Every all-even cycle type splits between live and dead operators. Among those types, the alive fraction falls as the number of cycles k increases: $[8]$ ($k=1$) is 83% alive, $[2\ 6]$ and $[4\ 4]$ ($k=2$) 70/67%, $[2\ 2\ 4]$ ($k=3$) 49%, and $[2\ 2\ 2\ 2]$ ($k=4$) 26%. Overall, 92.5% have a live genotype. The regime split is 82.7% live/live, 9.8% live/dead, 6.0% dead/dead, and 1.4% dead/live, so all four regimes occur among bijective operators.

Box 2: The rotation survey

Within the wrap-rotation subfamily, every non-trivial rotation is all-even and so possesses fixed rules (Supplement 2, Theorem 1), yet under the neighbour-blend coupling the tested rotations split by cycle structure (Supplement 3, Figure 1 and Figure 6):

- the single 8-cycles— $\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3 —*sustain* a high-diversity field, holding ~ 30 – 40 distinct rules through the longest runs we made (400 generations);
- offset ± 2 ($\gcd 2$, two 4-cycles) and offset ± 4 ($\gcd 4$, four 2-cycles) *collapse* to one or two rules within a few dozen generations.

The distinct-rule population curve (Figure 6) separates the two by more than an order of magnitude in a single scalar. Possessing a fixed rule and having the coupled field stay

near it are different properties. The present claim is limited to two seeds per rotation and treats population size as a descriptive observable, not as an order parameter in the statistical-mechanics sense.

Box 3: A braid of two collapsing operators can sustain the field

At width 80, horizon 120, and seed 0, offset +2 and offset +4 each have genotype change rate 0 when held alone, while their alternating braid has change rate 0.76 (Figure 8, top). The matched schedule control offset +2/offset -2 remains dead (Figure 8, bottom). This establishes that two collapsing components can construct a sustaining schedule. It does not establish the previously suggested block-system criterion: all three even offsets preserve the evens/odds partition, and both pairs generate the same translation subgroup $\langle +2 \rangle$. The cause of the contrasting braid outcomes remains to be identified.

Box 4: The mixing fraction is a reproducible diversity control

Let σ be displayed-position offset +1. We ran periodic 12-step schedules mixing the sustaining 8-cycle σ^1 with the collapsing two-4-cycle operator σ^2 , at width 80 for 120 generations and five seeds per mixture. Mean late-window diversity rises from 1.0 rule when the σ^1 fraction is zero, through 16.8 at fraction 0.25, to 36.9 at fraction 5/12; above that it remains in the range 34–41 rather than increasing monotonically (Figure 9). The schedule fraction is thus a tunable control for entry into a sustained high-diversity regime, with saturation rather than a continuously linear response.

Box 5: The rotation split occurs with and without neighbour blending

Figure 5 uses stochastic elementary writes without neighbour blending; Supplement 3, Figure 1 and Figure 6 include the blend. Over the displayed horizons, offset +1 remains active and offset +2 settles under both protocols, although their diversity levels and transients differ. These runs therefore do not identify neighbour blending as the cause of the split; they show that fixed-byte existence alone does not determine finite-horizon settling under either protocol.

Box 6: No finite-size evidence of a critical point over the tested range

The natural control parameter is the *propagator fraction* q ($q = 0$ all blend, $q = 1$ all propagator, offset +2). If the sustained phase sat at a critical point in q , a finite-size scan would show three things as the lattice grows: the transition in genotype activity $a_G(q)$ *sharpening*, its width shrinking toward zero; a susceptibility proxy peaking at a *fixed* interior value of q , the peak growing without bound; and Binder cumulants for different L crossing at that same value. Across widths $L = 30\text{--}240$ (32 seeds each) we observe none of the three.

First, $a_G(q)$ rises *smoothly*, well described by a logistic crossover ($R^2 > 0.99$, Figure 10a), and its width does not shrink over the tested sizes ($w \approx 0.13$, $w \sim L^{-0.04}$)—the transition is not sharpening. Second, the size-scaled run-to-run spread $L \text{Var}(a_G)$ (Figure 10b), used as a finite-size susceptibility proxy, does *peak* inside the crossover band and does grow ($\sim 5.7\times$ from the smallest width to the largest), but its location drifts rather than converging ($q = 0.75, 0.50, 1.00, 1.00$ as L increases), and for the two largest sizes the maximum lies at the sampled boundary $q = 1$, so no interior peak is resolved at all. Third, there is no Binder-cumulant crossing. The scan therefore provides no finite-size evidence for a sharpening transition over $L = 30\text{--}240$. These results support a broad crossover, plausibly influenced by seed-dependent residence times, rather than a confirmed critical point; they do not exclude different scaling outside the tested range.

Box 7: The river sustains structured phenotype

In all six tested seeds at $L = 80$, $T = 120$, the river produces coherent bands of alternating phenotype that drift through a disordered background. In the larger pinned run at $L = 240$, $T = 360$ (Figure 11), those structures persist while branching, merging, and reorganising across the full spacetime field. This is a bounded six-seed observation plus one larger representative run, not a claim that the structure is an attractor or survives arbitrarily long.

Box 8: Phenotype-to-genotype feedback roughly doubles causal reach

All runs use $L = 80$, $T = 120$, $t^* = 60$, with damage measured on the *phenotype* layer as the number of cells differing between branches and distance as the shorter circular displacement. At $dt \leq 10$ the two branches are empirically indistinguishable. By $dt = 59$, in a single-seed sweep over all 80 sites, the live construction has damaged 12.97 cells against the control's 5.51 and spread 16.95 against 7.46 (Figure 12 plots that sweep). Replication uses a separate design—six seeds, 20 evenly spaced sites each, aggregated to one mean per seed so that the *seed* is the unit of replication. The live construction exceeds its control in 6/6 seeds, per-seed differences +1.90, +7.95, +6.65, +6.60, +6.05, +0.85 cells, mean +5.00 (sd 2.90). Because $n = 6$ makes a t -statistic fragile we quote the exact sign-flip permutation test over all 2^6 assignments: the observed mean is the extremum, one-sided $p = 1/64 \approx 0.016$ (paired $t(5) = 4.23$ for comparison). Perturbations do not spread uniformly: in the single-seed sweep 22 of 80 sites are dead ends, and they form a contiguous region.

One qualification on what the control cuts. `run-river-ablated-from` reads a phenotype frozen at the start of each run segment, so it cuts the *dynamic* edge but still presents the genotype with a static field that, in the perturbed branch, already contains the flip. The 5.51 is therefore feedback-frozen, not feedback-absent; with the frozen reference supplied externally and identical in both branches, reach falls further, to 1.28 (Box 11). The doubling quoted here is the effect of making the read *current*, which is the smaller half of the total coupling effect.

Box 9: Causal reach on a scale calibrated against elementary rules

On a third design—four seeds, ten sites each, damage recorded at every dt —width grows as $dt^{0.64}$ for the live construction against $dt^{0.43}$ for its control, with centroid displacement scaling as $dt^{0.56}$ against $dt^{0.43}$. Exponents are least-squares fits over $dt \in [5, 59]$ on site-averaged curves, conditional on damage existing—a run whose perturbation has died contributes no width and is excluded rather than entered as zero—and we quote no

interval on them (Figure 14); these are reported descriptively and we draw no inference from them about gliders or particles, for a reason given below. Placed on one scale against elementary rules run under an identical protocol (mean damaged cells at $dt = 59$): rule 0 0.00; rule 204 1.00; rule 90 8.00; **the construction** 12.97; rule 54 15.05; rule 110 20.23; rule 30 37.50. The instrument returns exactly zero where nothing propagates, and the construction propagates at a magnitude comparable to the elementary rules that do.

Box 10: The order parameter is the strength of the phenotype-to-genotype coupling

Run under the protocol above— $L = 80$, $t^* = 60$, one phenotype flip, damaged phenotype cells at $dt = 59$ —the constructions separate by *coupling*, not by which parameter they vary. Where the genotype update never reads the phenotype, nothing clears the bottom of the complex band: the bare operators (P_a 0.03, rot+1 2.05), braided writings (0.12, 5.20), spatial niches (0.50, 0.72), asynchronous refuges (0.47, 0.75), replacement mutation (0.23, 0.85), the preserving limit (1.10), and blend strength across its whole range—1.00 giving 0.03, 0.70 giving 0.55, 0.35 giving 3.38, and 0.00, the most propagating feedforward configuration we found, giving 8.15, level with rule 90. Most sit below rule 204 (1.00), the frozen identity.

The coupling is not merely present or absent; it is a dial, and reach is graded and monotone along it. Conservative transport whose swap probability is gated on the local phenotype interface rises from 1.10 at rate 0 through 4.25 at 0.10, 9.43 at 0.20, 14.45 at 0.35, 19.27 at 0.50 and 26.05 at 0.75, crossing rule 90 between rates 0.10 and 0.20 and passing rules 110 and 54 before saturating. The river construction sits at 12.97 on the same scale, and cutting its single edge returns it to 5.51, back among the feedforward constructions.

That this is the *coupling* and not merely rule mobility is established by an ungated control. Replacing the phenotype-gated swap probability with a constant one leaves the mean number of swaps unchanged—for a balanced field the gated mean is exactly the rate—and leaves the dynamics bijective and the rule histogram invariant, so the two differ only in whether the genotype can see the phenotype. The ungated control never leaves the ordered band: 1.65, 1.68, 1.77, 2.77 and 4.05 at rates 0.20 through 1.00, against 9.43 to 25.38 for the gated version, a factor of 5.7 to 10.9 at matched mobility (Figure 13).

An order parameter therefore exists, and the classical picture is recovered in it: reach is a graded, monotone function of coupling strength with a locatable crossing into the complex band. What is unusual is only *which* coordinate it is. Every parameter internal to the update— λ , propagator fraction, sustained diversity, blend strength, mutation rate, refuge probability, niche width, and rule mobility itself—is silent or nearly so. The coordinate that governs is the gain of the loop from the phenotype back to the genotype, which conventional scans do not sweep because architecture is usually treated as a property a model has or lacks rather than as a quantity.

Box 11: Coupling is a dial, not a switch, and reach is monotone along it

Both constructions that read the phenotype admit a gain parameter, and in both causal reach is graded and monotone in it under the protocol of Box 9.

Figure 15 plots both dials. For conservative transport the gain is the phenotype-gated swap rate. Reach rises 1.10, 2.17, 4.25, 9.43, 14.45, 19.27, 26.05, 25.38 at rates 0, 0.05, 0.10, 0.20, 0.35, 0.50, 0.75, 1.00: it crosses rule 90 between rates 0.10 and 0.20, passes rules 110 and 54, and saturates past 0.75.

For the river the gain is the *currency* of the four context bits the genotype step reads. Each cell at each step reads the live phenotype with probability γ and a frozen reference otherwise, so γ is the fraction of the genotype's view of the phenotype that is causally

current. Over three seeds and eighty sites, reach rises 1.28, 3.00, 3.85, 3.43, 4.63, 5.00, 5.77, 7.38, 12.39 across $\gamma = 0$ to 1 in eighths, monotone within one standard error, with standard errors 0.19 to 0.57.

Three properties make the river dial an instrument rather than a knob. It is anchored at both ends: $\gamma = 1$ reproduces the live river at 12.97 exactly. The random tape is preserved—the propagator draw is consumed once per cell at every γ , with gate coins from a separate stream re-seeded identically in both branches, so the gate injects no divergence of its own. And because the frozen field is a real phenotype, marginals and spatial structure are matched at every γ ; only causal currency varies.

The two curves have opposite curvature. Transport is concave, buying most of its reach in the first fifth of the dial. The river is convex: 45% of its span arrives in the last eighth, and only $\gamma = 1$ clears rule 90, so one stale read in eight costs 40% of the reach. A plausible reading is that the river's first-match fires on a four-bit conjunction, so a stale context does not degrade the match but breaks it; we have not tested this, and gating the four bits independently rather than together would separate it.

A one-seed pilot showed a non-monotonic dip near $\gamma = 0.5$ that did not survive three seeds. We record it because we nearly believed it.

Box 12: Sustained diversity does not determine causal reach; conservative transport breaks the apparent interior optimum

The original 76 configurations produce an apparent rise and fall: binned medians rise from 0.10 cells below 40 sustained rules to 2.34 at 125–135, then fall to zero above 135. That descent is completely confounded. Every original configuration above 135 obtains its diversity either by replacement mutation or by holding the genotype fixed. The observed failures are real—mutation erases the lesion and holding prevents its spread—but they do not show that high diversity suppresses propagation.

We therefore add a conservative eighth mechanism. Each run begins from a balanced empirical rule measure with support 32, 64, 96, 128, 160, 192, 224, or 256. Alternating layers of disjoint local pairs exchange their rules with a probability controlled by a transport rate and the local phenotype interface. These exchanges are bijections: no rule is created or destroyed, and the complete genotype histogram and its Shannon diversity remain invariant at every checked step. A local transposition supplies the perturbation while preserving that same measure in both branches. We run five transport rates, six seeds, and eight perturbation sites for 40 response steps, giving 1,920 paired responses. Figure 16 reports causal amplification—final genotype damage divided by the two initially exchanged sites—so the conservative responses share a scale with the original single-site survey. With transport zero, amplification is exactly 1.00 at every diversity. At transport rates 0.25, 0.50, 0.75, and 1.00, the ranges across the full diversity axis are respectively 6.63–8.26, 14.11–15.31, 18.75–21.26, and 20.78–22.08. There is no descending limb: within each transport rate the dependence on diversity is approximately flat or slightly positive, while changing transport changes reach by more than an order of magnitude.

Sustained rule diversity is therefore not an order parameter for causal reach. Mutation and holding are distinct suppression mechanisms, but they are not flanks of a common diversity optimum. Structured genotype transport is the control exposed by this experiment.